

Multimodal Fusion for Parkinson’s Disease Detection using Speech and Motor Tapping Data

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TABLE I
PATIENT- AND SESSION-LEVEL DISTRIBUTION OF THE mPOWER DATASET
FOR EACH MODALITY.

	Audio		Tapping	
	PD	Healthy	PD	Healthy
Number of patients	302	230	302	230
Number of sessions	11,852	3,092	12,387	3,378
Min sessions per patient	1	1	1	1
Max sessions per patient	513	383	512	389

Abstract—Parkinson’s disease (PD) is associated with motor and speech impairments that can be captured using digital biomarkers. We address binary classification of PD patients versus healthy controls using sustained phonation audio and finger-tapping data from the mPower dataset. We evaluate unimodal models for each modality and multimodal fusion approaches at the patient level.

I. INTRODUCTION

Parkinson’s disease (PD) is a neurodegenerative disorder characterized by motor impairments and altered speech production. Recent advances in mobile technologies enable large-scale collection of digital biomarkers reflecting these symptoms, but integrating heterogeneous modalities remains challenging.

In this work, we investigate whether combining speech and motor digital biomarkers can improve Parkinson’s disease detection. We first evaluate unimodal models for each modality independently and then assess multimodal fusion strategies at the patient level.

Our contributions are:

- unimodal classification models for audio based on acoustic features, spectrogram representations and raw audio time-series;
- unimodal classification models for finger-tapping data using temporal-spatial features and spatial heatmaps;
- a comparison of early, intermediate, and late multimodal fusion strategies to evaluate the benefit of integrating audio and motor modalities.

II. DATASET AND TASK DEFINITION

A. Dataset

We use the mPower dataset, consisting of remote assessments collected from Parkinson’s disease patients and healthy controls via a mobile application. The original dataset is imbalanced at the patient level, and participants may contribute a variable number of sessions. Two modalities are considered: sustained phonation audio recordings and finger-tapping interaction data with timestamped button events and spatial coordinates. The classification task is binary, distinguishing PD patients from non-PD controls.

Table I reports the distribution of patients and sessions for each class.

B. Evaluation Protocol

Since participants may contribute multiple recording sessions, predictions are generated at the session level and aggregated to produce a single patient-level label (e.g., via majority voting), ensuring evaluation at the patient level. Model performance is assessed using the Area Under the ROC Curve (AUC), a threshold-independent metric that measures discrimination between Parkinson’s disease and control patients and is robust to variations in class prevalence.

III. METHODS

An overview of the full pipeline, from raw data to patient-level predictions, is shown in Figure 1.

A. Folds and Early Stopping

Patients were split into 5 disjoint folds, with each patient assigned exclusively to the training, validation, or test set, and all sessions from a given patient kept within the same split. Models were trained on patient-balanced training sets, while validation and test sets remained imbalanced to reflect realistic class distributions. Early stopping was based on validation AUC, and final performance was assessed on the test set using mean AUC across folds. Session-level class imbalance due to unequal numbers of sessions per patient was handled during training (Section III-C).

B. Data Preprocessing

1) *Audio preprocessing*: Audio data consist of 10-second sustained phonation recordings (“Ahh”) provided as .m4a files. Recordings that were too short (three in total) are excluded. The audio is converted to .wav format, silence-trimmed, normalized, and stored as one-dimensional .npy waveforms. Mel spectrograms are obtained using a predefined pipeline. From the normalized waveforms, acoustic features are extracted, including MFCCs and Delta coefficients, spectral descriptors, voice quality measures, and energy-related features. All features are computed and stored at the session level in tabular (.csv) format.

2) *Tapping preprocessing*: Finger-tapping data are provided as JSON files containing timestamped tap events, button identifiers, and tap coordinates. Invalid entries and incomplete sessions are removed. For each session, inter-tap intervals (Δt), button usage statistics, and spatial coordinates are extracted. Session-level representations are computed independently and stored at the session level in a tabular (.csv) format.

C. Class Imbalance Handling and Aggregation Strategy

For both audio and tapping models, session-level class imbalance is handled using class-weighted loss, undersampling of the majority class, or balanced mini-batch sampling via a `WeightedRandomSampler`. These strategies are compared during model selection (see Section IV). Session-level

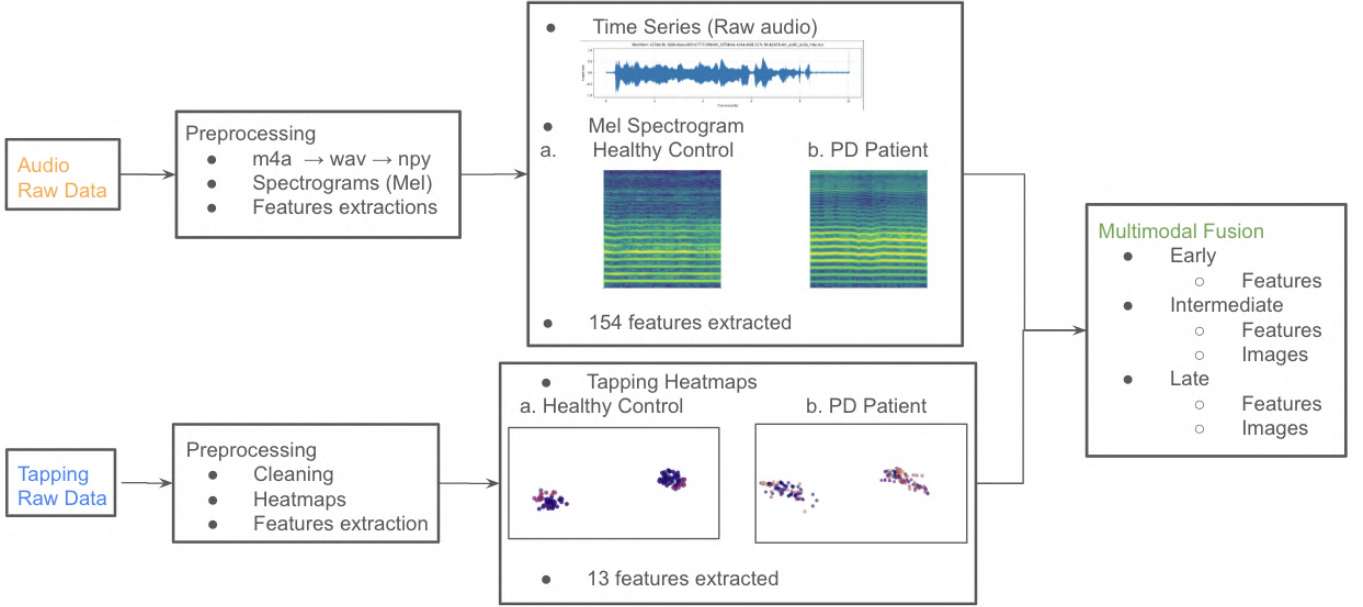


Fig. 1. Overview of the multimodal pipeline. Audio and finger-tapping data are preprocessed independently, evaluated using unimodal models with different representations, and combined using multiple multimodal fusion strategies.

predictions are aggregated to the patient level using either mean probability averaging or majority voting.

D. Optimization and Training Strategy

All models are trained using the Adam optimizer with cross-entropy loss. Learning rates are dynamically adjusted using a ReduceLROnPlateau scheduler based on validation AUC to ensure stable convergence.

E. Audio Unimodal Models

For the audio modality, three representations are evaluated: a feature-based representation, a time–frequency representation, and a one-dimensional temporal representation of the raw waveform (Figure 1).

1) *Feature-based representation*: Session-level acoustic features are extracted and classified using a multilayer perceptron (MLP). Two feature sets are considered: V1, capturing core spectral, pitch, voice quality, and energy characteristics, and V4, extending these descriptors with temporal dynamics and higher-order acoustic information. For all features, mean and standard deviation statistics are computed over the session.

The full feature vector is fed into an MLP composed of successive Linear–ReLU–Dropout blocks, a final hidden layer of size 16, and a linear output layer producing two logits for binary classification (PD vs Healthy). Due to the high input dimensionality, architectures with an additional hidden layer are also explored. A one-dimensional ResNet architecture, as well as alternative optimizers and loss functions, were evaluated but did not yield competitive performance and are therefore omitted.

2) *Spectrogram-based representation*: Audio signals are also represented using precomputed Mel spectrograms. Linear spectrograms are initially considered but discarded in favor of Mel representations, which have shown superior performance in CNN-based audio classification [3, 2].

Mel spectrograms are classified using convolutional neural networks. A custom 2D CNN and a ResNet-based model are evaluated, but the most effective approach relies on an Inception V3 architecture pre-trained on ImageNet, previously

shown to perform well for Parkinson’s disease detection from voice data [3]. The backbone is kept frozen, with tuning performed on the classification head through variations in optimization settings and trainable layers.

3) *Temporal waveform representation*: Normalized audio waveforms are evaluated using a Temporal Convolutional Network (TCN). Each session is represented as a multichannel time series, segmented into overlapping windows during training. The TCN consists of stacked temporal blocks with dilated one-dimensional convolutions and residual connections, enabling modeling of long-range temporal dependencies directly in the time domain.

F. Tapping Unimodal Models

For finger-tapping data, we evaluate two representations: a feature-based representation and a spatial heatmap representation (as shown in Figure 1).

1) *Feature-based representation*: Tapping behavior is represented using session-level statistical features and evaluated with a multilayer perceptron (MLP). Two groups of features are extracted. The first group consists of basic features summarizing motor timing, spatial dispersion, and task execution, including statistics of inter-tap intervals, tap counts, tap coordinates, and missed taps. The second group captures temporal dynamics and motor variability, including fatigue-related trends, variability indices, inter-tap distance statistics, and fluctuation measures.

The combined feature set is used as input to an MLP composed of successive Linear–ReLU–Dropout blocks with configurable dimensions, a final hidden layer of size 16, and a linear output layer producing two logits for binary classification (PD Vs Healthy). Models are evaluated using basic features only, advanced features only, and their combination.

2) *Spatial Heatmap Representation*: Alternatively, tapping trials are represented as spatial heatmaps, where tap locations are projected onto a 2D plane and colored according to inter-tap intervals. Spatial coordinates and timing values are normalized using global percentile-based scaling to ensure consistency across sessions. Heatmap images are classified

using a 2D convolutional neural network composed of convolutional blocks followed by global average pooling and a fully connected classification head.

G. Multimodal Fusion

Audio Data and Tapping Data were combined at different levels in an attempt to improve the results of the unimodal classification results [1]. It corresponds to the final stages of the pipeline shown in Figure 1.

1) *Early Fusion*: To ensure a consistent number of paired sessions per patient across modalities, sessions were randomly subsampled when necessary. The resulting sessions were then concatenated patient-wise, with session order randomized. The fused extracted features were then used to train various MLP architectures and tunings, which were compared as discussed in Subsection III-A.

2) *Intermediate Fusion*: Intermediate fusion was performed on the extracted features from audio and tapping data, as well as on the audio spectrograms and tapping heatmaps. In each case, one-dimensional embeddings were extracted from the different sources of data and concatenated before being passed through a final MLP. The embedding extractor models for the extracted features were MLPs and were trained simultaneously alongside the final MLP in the pipeline. On the other hand, image embeddings were extracted using the pre-trained ResNet18 model from Timm [4], which is specifically designed to extract one-dimensional image embeddings of length 512.

3) *Late Fusion*: Late fusion leverages the outputs of previously developed unimodal classifiers trained on audio and tapping extracted features. Specifically, MLP models generate session-level predictions for audio and tapping data for the test patients in each cross-validation fold. These predictions are then aggregated to obtain patient-level class decisions using late fusion strategies, including weighted averaging of class probabilities or predicted labels, by assigning greater weight to the tapping modality, which emerged as the most informative data source, and overall majority voting. This aggregation is performed independently for the test set of each fold.

IV. RESULTS

A. Audio Unimodal Results

We evaluate unimodal audio models using feature-based, spectrogram-based, and temporal waveform representations, with performance reported at the patient level using 5-fold cross-validation. Temporal waveform models exhibit poor performance and are not further considered.

The feature-based MLP achieves the strongest audio-only performance, outperforming spectrogram-based CNNs. The best configuration uses all 154 acoustic features, undersampling for class imbalance, and mean probability aggregation, with a learning rate of 5×10^{-4} , weight decay of 1×10^{-4} , dropout rate of 0.5, and hidden layer dimensions of 32 and 64. However, performance remains modest overall, indicating limited discriminative power of audio features alone.

Among spectrogram-based approaches, the Inception architecture performs best, although still below the feature-based MLP, and is retained for multimodal fusion. Alternative optimization strategies do not yield consistent improvements. Table II summarizes the impact of class imbalance handling, reporting AUC values using the best aggregation method for each model. Overall, unimodal audio models remain less discriminative than tapping-based approaches.

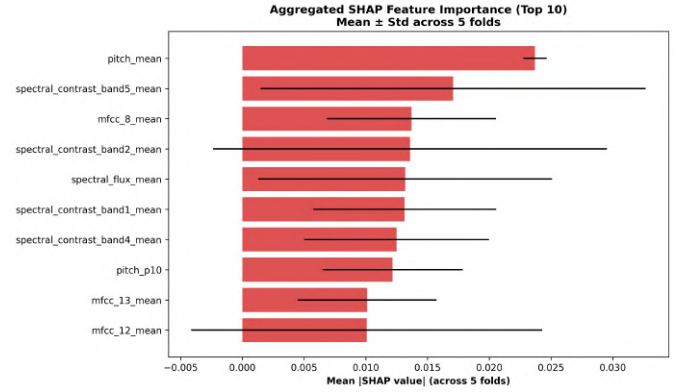


Fig. 2. Feature Contributions to the Audio-Based MLP Predictions (SHAP Analysis)

B. Tapping Unimodal Results

We evaluate unimodal finger-tapping models using feature-based and spatial heatmap representations, with results reported at the patient level.

Feature-based MLP models consistently outperform CNNs trained on spatial heatmaps. Combining basic and advanced tapping features yields the best performance, with additional gains from explicit class imbalance handling. The best tapping model is an MLP trained on the combined feature set using balanced mini-batch sampling via a WeightedRandomSampler, with patient-level aggregation based on mean probability averaging. This configuration achieves the highest patient-level AUC among all tapping-based models, with optimal hyperparameters: learning rate 5×10^{-4} , weight decay 5×10^{-3} , dropout rate 0.2, and hidden layer dimensions of 32 and 16.

In contrast, CNNs trained on spatial heatmaps exhibit lower performance but is still retained for subsequent multimodal fusion experiments. Table II summarizes the effect of different class imbalance handling strategies.

C. Effect of Aggregation Strategy

Mean probability averaging and majority voting yield comparable performance for both modalities. Therefore, mean probability averaging is adopted as the default aggregation strategy for consistency.

D. Feature Importance Analysis

Feature importance is analyzed using SHAP values computed for the best-performing MLP models with optimal hyperparameters. SHAP values quantify each feature's contribution to the predicted class probability.

For the audio modality (Figure 2), aggregated SHAP values highlight pitch-related features, spectral contrast, and MFCC descriptors as the main contributors to the model's predictions.

For the tapping modality (Figure 3), aggregated SHAP values highlight missed_taps, spatial variability (e.g., std_y, mean_y), and timing-related features as the main contributors, reflecting altered motor control in Parkinson's disease.

E. Multimodal Fusion Results

1) *Early Fusion*: The best early fusion MLP achieved a mean test AUC score of 0.75, as shown by Table III. On the one hand, it outperformed the best obtained MLP from the audio extracted features, which displayed a considerably lower test AUC score of 0.69. However, it failed to improve

TABLE II
PATIENT-LEVEL AUC FOR AUDIO- AND TAPPING-BASED MODELS UNDER DIFFERENT CLASS IMBALANCE HANDLING STRATEGIES.

Modality	Model	Baseline	Undersampling	Class Weights	Weighted Random Sampler
Audio	MLP (features)	0.685	0.686	0.685	0.6836
	Inception (MelSpec)	0.637	0.644	0.640	0.641
Tapping	MLP (features)	0.75	0.76	0.73	0.78
	CNN (heatmaps)	0.63	0.70	0.65	0.65

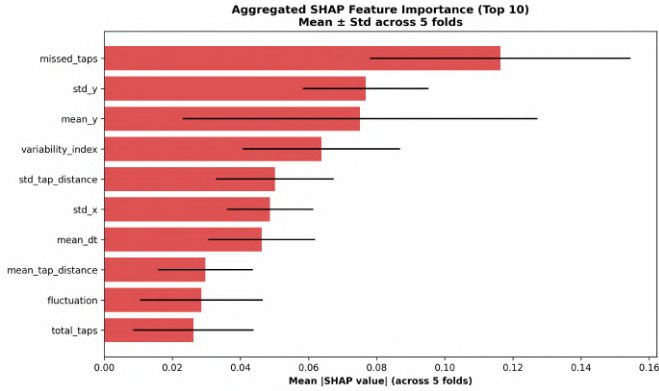


Fig. 3. Feature Contributions to the Tapping-Based MLP Predictions (SHAP Analysis)

TABLE III
TEST METRICS OF THE DIFFERENT FUSION STRATEGIES

Fusion	Early	Intermediate		Late
	Features	Features	Images	Features
AUC	0.75	0.74	0.76	0.77
F1	0.56	0.71	0.58	0.76
Accuracy	0.62	0.69	0.64	0.72

on the best MLP from the tapping extracted features, which demonstrated the highest observed mean test AUC score, namely 0.78 (cf. Table II). Of note, assigning class weights in MLP’s loss function consistently displayed better test metrics than other balancing techniques at the session level.

2) *Intermediate Fusion*: Intermediate fusion was applied to both extracted features (audio and tapping) and image-based representations (spectrograms and heatmaps). In terms of AUC, the image-based intermediate fusion outperformed the feature-based approach (0.76 vs. 0.74), while the opposite trend was observed for Accuracy and F1 score (cf. Table III).

Image-based intermediate fusion substantially improved upon the unimodal image classifiers: the best unimodal image model achieved an AUC of 0.65, compared to 0.76 for the intermediate fusion model, as shown in Table III. In contrast, intermediate fusion using extracted features improved over audio-only models but achieved performance comparable to tapping-only unimodal models.

Overall, intermediate fusion consistently yielded better test performance than early fusion.

3) *Late Fusion*: Late fusion combined the unimodal audio and tapping classifiers by integrating their extracted feature-based predictions. Individually, the audio and tapping models achieved baseline AUC scores of 0.67 and 0.74, respectively. In contrast, the late fusion approach substantially improved performance, reaching an AUC of 0.77, as reported in Table III. Similar gains were observed for Accuracy and F1 score: the unimodal audio and tapping classifiers obtained

Accuracies of 0.63 and 0.69, respectively, which increased to 0.72 after fusion, while their F1 scores (0.65 for audio and 0.72 for tapping) improved to 0.76. The best late fusion performance was obtained by computing weighted averages of the class probabilities. Nevertheless, alternative late fusion strategies based on weighted averaging of predicted labels and majority voting yielded comparable results, consistently outperforming the unimodal models.

V. DISCUSSION

Across fusion strategies, a consistent pattern emerges: early and intermediate fusion improve performance relative to weaker unimodal baselines, particularly audio-only models, but fail to surpass the strongest unimodal classifier based on tapping features. In terms of AUC, both approaches remain bounded by the performance ceiling set by the feature-based tapping MLP, suggesting that combining modalities at the input or representation level does not yield additional discriminative power beyond the most informative modality. The superior performance of tapping-based models may partly reflect the higher noise and variability present in the audio recordings, which can degrade the reliability of acoustic features.

Late fusion is the only strategy that consistently improves performance over unimodal baselines, yielding gains in AUC, Accuracy, and F1 score by integrating the outputs of independently optimized classifiers. Despite these improvements, the feature-based tapping MLP remains the best-performing overall model in this study. More broadly, these findings indicate that multimodal fusion does not inherently outperform strong unimodal models; its effectiveness depends on both the fusion strategy and the relative informativeness of the modalities.

The consistently moderate performance across models (AUC < 0.8), despite the use of class imbalance mitigation techniques, suggests that handling session imbalance remains a key challenge. Ultimately, late fusion provides the most reliable performance gains compared to unimodal classifiers, while tapping-based models remain the most effective standalone solution for Parkinson’s disease detection.

VI. CONCLUSION

To conclude, unimodal audio-based feature models led to limited discrimination between Parkinson’s patients and healthy controls. Conversely, unimodal tapping-based feature models and multimodal classifiers proved more effective. Still, in a stringent medical context, the presented classifiers would, unfortunately, prove inefficient compared to existing clinical approaches. Other studies have identified models with considerably higher AUC scores [3], motivating further research.

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ETHICAL RISKS

The mPower dataset used in this project is governed by a rigorous and participant-centred data governance framework designed to balance scientific openness with strong privacy protections. Data were donated voluntarily by thousands of participants through a remote enrollment process that included a visually guided and robust informed consent procedure. Crucially, participants were explicitly asked whether their data could be used for secondary research, and only data from individuals who consented to broad secondary use are included in the dataset analyzed here. This opt-in mechanism ensures that participants retain agency over the downstream use of their sensitive health data. In addition, access to the mPower dataset is subject to formal governance structures that regulate data sharing and enforce privacy safeguards, reflecting the novel nature of the remote data collection process. Together, these measures substantially mitigate ethical risks related to data misuse, lack of informed consent, or unauthorized access.

Despite these protections, ethical risks arise from the reliability of the utilized data and the performance limitations of our machine learning models. Firstly, the remotely generated data by the patients could have been falsified. For example, any individual could have performed a tapping trial by tapping randomly all around the screen, corrupting the reliability of the trial. In this dataset’s context, we cannot verify whether this was the case, as the process of data collection was fully remote. Secondly, the relatively low Accuracy, F1 Score, and AUC indicate a non-negligible risk of misclassification: either labelling healthy individuals as Parkinson’s patients or, more critically, failing to identify actual patients. Such errors could have serious consequences if the model were misinterpreted as a diagnostic tool rather than a research prototype. Additionally, all models were trained using the given fixed 5-fold cross-validation splits, with validation performed after each epoch on the same validation sets and testing conducted on identical test sets per fold. This setup introduces a risk of implicit model selection bias, as certain models may appear superior simply because they fit the provided test data better, rather than exhibiting true generalization capability. To mitigate this, we maintain full transparency regarding our training, validation, and testing procedures and explicitly report all performance limitations. However, further mitigation is constrained by the absence of additional independent test data. Access to supplementary data would have allowed for shuffling, re-splitting, or external validation, thereby improving the statistical robustness and ethical reliability of our conclusions.

A. Additional Materials

This appendix contains supplementary figures and analyses that provide additional insight into the models' behavior and feature contributions. These materials are included to support the main results presented in the paper, but are not essential to the core discussion.

B. Hyperparameter Search Space

This part reports the hyperparameter grids used during model selection for the multilayer perceptron (MLP) models trained on extracted features. Separate grids were defined for the tapping and audio modalities to account for differences in feature dimensionality and model capacity.

1) *Tapping Feature MLP Hyperparameters*: Table IV summarizes the hyperparameter grid used for MLP models trained on tapping-based extracted features.

TABLE IV
HYPERPARAMETER GRID FOR THE TAPPING FEATURE-BASED MLP MODELS.

Hyperparameter	Values
Learning rate	{0.001, 0.0005}
Weight decay	$\{1e^{-4}, 1e^{-3}, 5e^{-3}\}$
Dropout rate	{0.2, 0.5, 0.7}
Hidden layer 1 size	{32, 64, 128}
Hidden layer 2 size	{16, 32, 64}

2) *Audio Feature MLP Hyperparameters*: Table V summarizes the hyperparameter grid used for MLP models trained on audio-based extracted features.

TABLE V
HYPERPARAMETER GRID FOR THE AUDIO FEATURE-BASED MLP MODELS. AN OPTIONAL THIRD HIDDEN LAYER WAS EVALUATED TO ASSESS WHETHER INCREASED MODEL DEPTH IMPROVED PERFORMANCE.

Hyperparameter	Values
Learning rate	{0.001, 0.0005, 0.0001}
Weight decay	$\{1e^{-4}, 5e^{-4}, 1e^{-3}\}$
Dropout rate	{0.3, 0.5, 0.7}
Hidden layer 1 size	{256, 128}
Hidden layer 2 size	{128, 64}
Hidden layer 3 size (optional)	{64, 32}

C. Per-Fold SHAP Feature Importance

This section presents non-aggregated SHAP feature importance plots for each cross-validation fold, separately for the audio and tapping modalities, to illustrate feature priority variability across folds. Visual inspection of the SHAP distributions shows that many audio features yield SHAP values close to zero for a large number of samples, resulting in relatively flat importance profiles and suggesting limited discriminative and probabilistic power for many individuals. In contrast, tapping features exhibit more structured and variable importance patterns across folds, indicating a stronger and more consistent contribution to model predictions.

1) *Audio Modality*: Non-aggregated SHAP feature importance across the five cross-validation folds is shown in Figure 4.

2) *Tapping Modality*: Non-aggregated SHAP feature importance across the five cross-validation folds is shown in Figure 5.

D. Grad-CAM Visualizations and Confusion Matrices

This section provides qualitative Grad-CAM visualizations illustrating how CNN-based models attend to relevant input regions, along with "confusion matrices" summarizing their classification behavior.

1) *Audio Spectrogram CNN*: Figure 6 (top panel) illustrates representative Grad-CAM triplets for the audio spectrogram CNN, showing that model attention is primarily concentrated on localized time–frequency regions of the spectrogram.

Figure 6 (bottom panel) presents Grad-CAM overlays for samples corresponding to each confusion matrix category, highlighting how differences in attended spectrogram regions may contribute to correct versus incorrect predictions. Correct predictions (TP and TN) tend to focus on stable harmonic structures, whereas misclassifications (FP and FN) exhibit more diffuse or shifted activation patterns, suggesting sensitivity to spectro-temporal variability.

2) *Tapping Heatmap CNN*: Figure 7 (top panel) shows representative Grad-CAM triplets for the tapping heatmap CNN, indicating that the model primarily attends to spatial clusters of tap locations.

Figure 7 (bottom panel) presents Grad-CAM visualizations across confusion matrix outcomes, illustrating how spatial attention differs between predictions. In both cases predicted as PD (true and false positives), the model appears biased toward peripheral or corner regions of the input rather than the tapping clusters themselves, suggesting reliance on non-specific spatial cues. By contrast, in both cases predicted as HC (true and false negatives), attention is more clearly aligned with the tapping clusters, despite one resulting in a misclassification. This pattern may indicate that spatial tapping structure is more consistently captured for HC predictions, while PD predictions rely on less localized and potentially ambiguous features, possibly reflecting residual effects of class imbalance that bias the model toward PD predictions in uncertain cases.

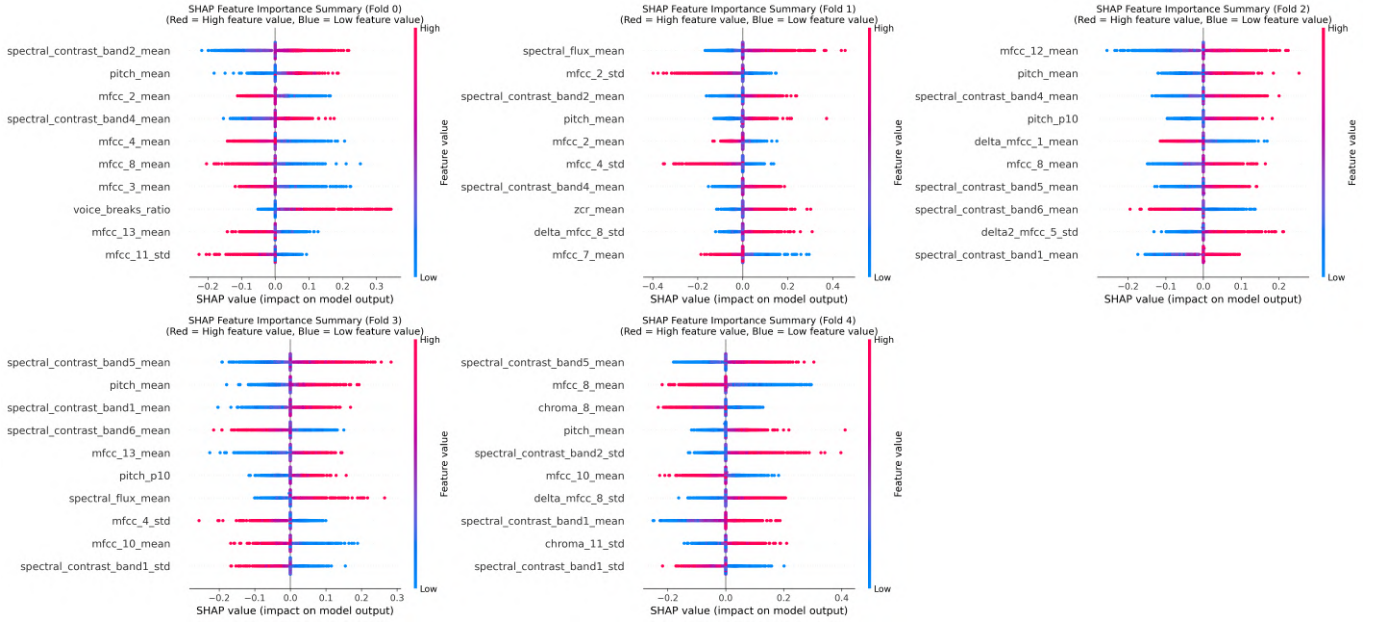


Fig. 4. Non-aggregated SHAP feature importance for the audio-based MLP across the 5 cross-validation folds.

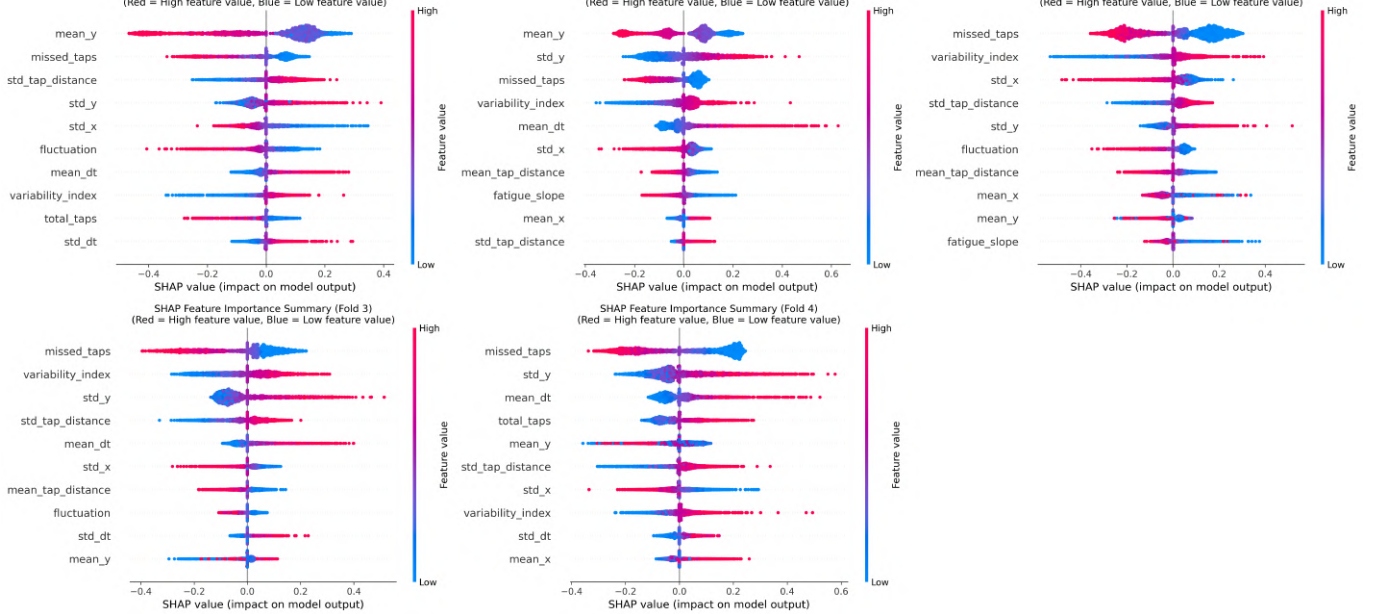
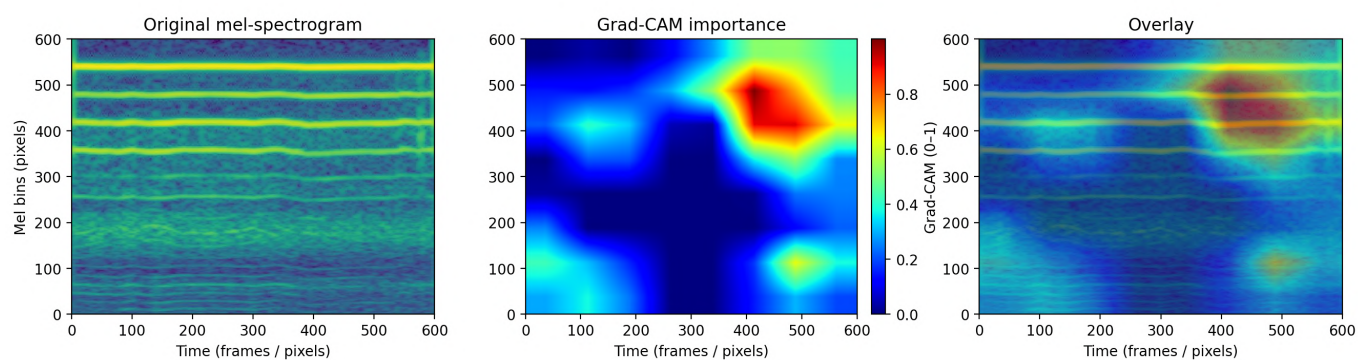


Fig. 5. Non-aggregated SHAP feature importance for the tapping-based MLP across the 5 cross-validation folds.

HealthCode: dc600bb6-ad90-42ee-99ee-d665ab1a62ba | Pred: PD ($p=0.66$) | True: PD



Grad-CAM overlays — Confusion Matrix

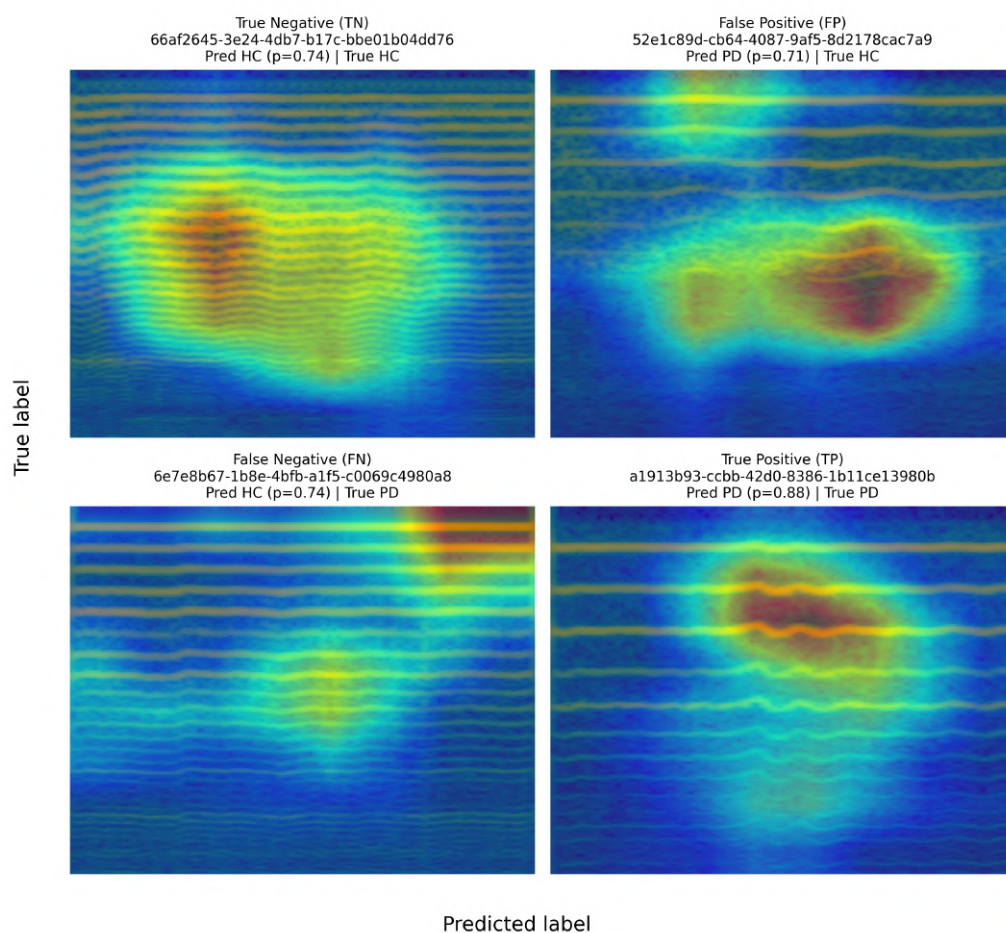
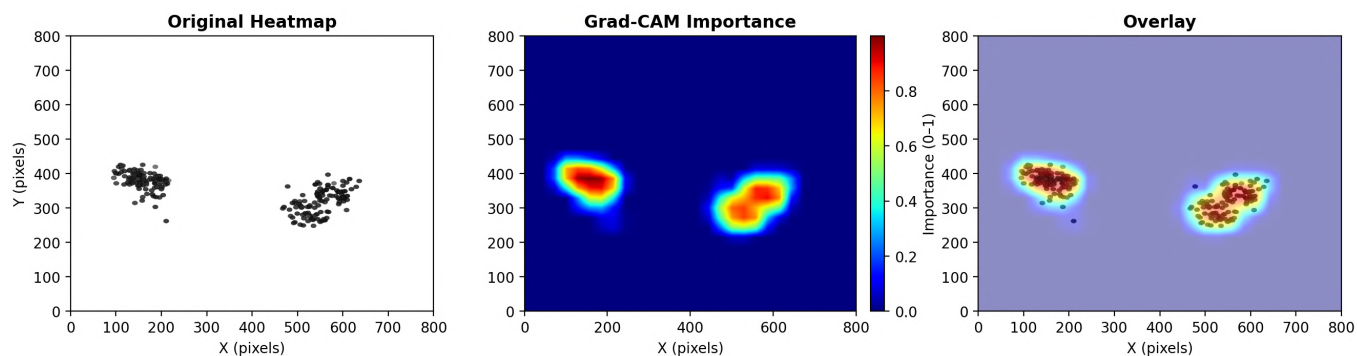


Fig. 6. Grad-CAM triplet and Grad-CAM overlays for confusion matrix cases for the audio spectrogram CNN.

HealthCode: 2b1f8af1-c928-4ca5-a240-fdd3ef6ead98 | Pred: HC (p=0.611) | True: PD

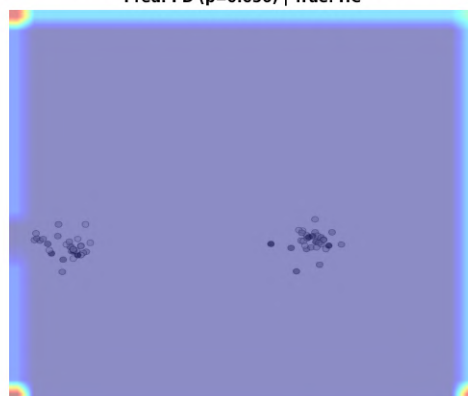


Grad-CAM Analysis: Confusion Matrix Cases

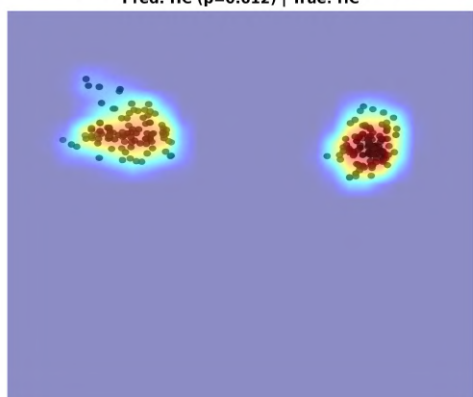
TP (True Positive)
faeab8f9-5e6b-48ef-8e87-3cc7e6da3e5f
Pred: PD (p=0.656) | True: PD



FP (False Positive)
909ff077-9ee3-467a-9e71-454394b0eb8a
Pred: PD (p=0.656) | True: HC



TN (True Negative)
617f9a7c-ce57-4bc1-876b-72f904220934
Pred: HC (p=0.612) | True: HC



FN (False Negative)
2b1f8af1-c928-4ca5-a240-fdd3ef6ead98
Pred: HC (p=0.639) | True: PD

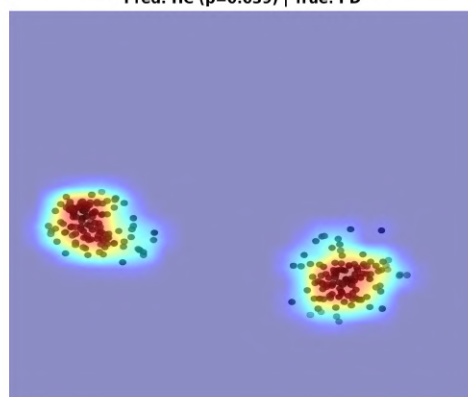


Fig. 7. Grad-CAM triplet and Grad-CAM overlays for confusion matrix cases for the tapping heatmap CNN.